QDArchive

SQLite Metadata Schema Proposal

`files` table — column definitions and rationale

Project	QDArchive Seeding Pipeline
Version	v1 (2026-03)
Institute	FAU Erlangen-Nürnberg
Table	`files`
Engine	SQLite 3 (WAL mode)

1. Design Rationale

The pipeline stores all metadata in a single **denormalised** table named `files`. Each row represents *one file* within a dataset. A dataset containing five files produces five rows that share the same `doi`, `title`, and `author` values but differ in `url`, `local_filename`, and `file_type`.

This design is intentional for the acquisition phase:

- **No joins required** during high-throughput ingestion.
- **Deduplication** is a single `SELECT url` lookup before every insert.
- **Resume support** is trivial — the already-seen URL set is loaded once at startup from the existing database.
- Normalisation into separate `datasets` and `files` tables can be applied in Part 2 with no data loss.

Metadata-only mode: When the pipeline runs without `--download`, `local_dir` is left empty but `local_filename` is still populated. This enables a fast inventory scan without consuming disk space. Files can be downloaded later by querying rows where `download_timestamp` = ''.

2. Column Definitions

The table has **23 columns** organised into six logical groups. ✓ marks columns that are always populated; – marks columns that may be empty depending on repository availability or download mode.

2.1. Identity

Column	Type	Req.	Source	Description
`id`	*INTEGER*	✓	auto	Auto-incrementing primary key.
`url`	*TEXT*	✓	all	Direct file download URL. Primary deduplication key: checked before every insert to prevent recording the same file twice across sessions.
`doi`	*TEXT*	–	all	Persistent dataset-level DOI, e.g. `10.5281/zenodo.14523891`. Links all files of the same dataset and enables cross-repository deduplication.

2.2. File

Column	Type	Req.	Source	Description
`local_dir`	*TEXT*	✓	auto	Relative path of the download folder from the archive root, e.g. `zenodo/14523891`. Empty in metadata-only mode.
`local_filename`	*TEXT*	✓	auto	Filename as provided by the repository, e.g. `nurse-patient-study.qdpx`. Populated in both metadata-only and download modes.
`file_type`	*TEXT*	✓	auto	Classification derived from the file extension: `analysis` (QDA project files), `primary` (transcripts, recordings, documents), or `additional` (README s, codebooks, images).
`file_extension`	*TEXT*	✓	auto	Lowercase extension including the leading dot, e.g. `.qdpx`. Stored separately to allow fast filtering without substring operations on the filename.
`file_size_bytes`	*INTEGER*	–	all	File size in bytes as reported by the repository API. Not always provided. Useful for storage planning and detecting truncated downloads.
`checksum_md5`	*TEXT*	–	auto	MD5 hex digest. Empty until the file is downloaded. Dryad supplies its own server-side digest; all other repositories compute it locally.

2.3. Download

Column	Type	Req.	Source	Description
`download_timest`	*TEXT*	✓	auto	ISO-8601 UTC timestamp set on successful download, e.g. `2026-03-10T14:32:05`. Empty string when the file has not been downloaded. Identifies pending downloads in queries.
`source_name`	*TEXT*	✓	auto	Human-readable repository name, e.g. `Zenodo`, `Dryad`, `Harvard Dataverse`.
`source_url`	*TEXT*	–	all	Dataset landing page URL (not the file URL). Allows any record to be traced back to its origin in a browser.

2.4. Dataset

Column	Type	Req.	Source	Description
`title`	*TEXT*	–	all	Title of the dataset as provided by the depositor.
`description`	*TEXT*	–	all	Abstract or description, truncated to 500 characters.
`year`	*TEXT*	–	all	Four-digit publication year (YYYY), extracted from the repository's publication date field.
`keywords`	*TEXT*	–	all	Semicolon-separated keywords as provided by the depositor, e.g. `NVivo; interview; thematic analysis`.
`language`	*TEXT*	–	all	Language of the dataset content. Zenodo and Dataverse expose this field; Dryad does not. Important for qualitative data usability.

2.5. People

Column	Type	Req.	Source	Description
`author`	*TEXT*	–	all	Semicolon-separated list of all credited author names as listed in the dataset metadata record.
`uploader_name`	*TEXT*	–	all	Name of the depositor. May differ from the listed authors. Extracted from Dataverse `datasetContact`, Zenodo `creators[0]`, or Dryad `authors[0]`.
`uploader_email`	*TEXT*	–	all	Contact email of the depositor. Useful for follow-up if files are corrupted, restricted, or require clarification.

2.6. Legal

Column	Type	Req.	Source	Description
`license`	*TEXT*	–	all	License short name or SPDX identifier, e.g. `CC0-1.0` or `CC-BY-4.0`. Only open-license datasets are ingested.
`license_url`	*TEXT*	–	all	URL pointing to the full license text.

2.7. Discovery

Column	Type	Req.	Source	Description
matched_query	*TEXT*	–	auto	The exact search query string that caused this dataset to appear in results, e.g. `.qdpx` or `NVivo` or `thematic analysis`. Enables post-hoc analysis of query productivity across tiers and repositories.

3. Indexes

Five indexes are created automatically on database initialisation to support the most common access patterns:

Index	Purpose
idx_url	Checked before every insert. Prevents duplicate URL entries across sessions.
idx_doi	Retrieves all files belonging to a dataset by DOI.
idx_file_type	Fast isolation of `analysis` files for Part 2 processing.
idx_source	Filters or aggregates results by repository name.
idx_matched_query	Analyses productivity of each search query tier.

4. CREATE TABLE Statement

```
CREATE TABLE IF NOT EXISTS files (
    id                  INTEGER  PRIMARY KEY AUTOINCREMENT,

    -- Identity
    url                 TEXT NOT NULL,    -- deduplication key
    doi                 TEXT,

    -- File
    local_dir           TEXT,             -- empty in metadata-only mode
    local_filename      TEXT,
    file_type           TEXT,             -- analysis | primary |
        additional
    file_extension      TEXT,
    file_size_bytes     INTEGER,
    checksum_md5        TEXT,

    -- Download
    download_timestamp  TEXT,             -- ISO-8601 UTC; empty if
        pending
    source_name         TEXT,
    source_url          TEXT,

    -- Dataset
    title               TEXT,
```

```sql
    description         TEXT,           -- truncated to 500 chars
    year                TEXT,           -- YYYY
    keywords            TEXT,           -- semicolon-separated
    language            TEXT,

    -- People
    author              TEXT,           -- semicolon-separated
    uploader_name       TEXT,
    uploader_email      TEXT,

    -- Legal
    license             TEXT,           -- SPDX id or short name
    license_url         TEXT,

    -- Discovery
    matched_query       TEXT
);

CREATE INDEX IF NOT EXISTS idx_url           ON files(url);
CREATE INDEX IF NOT EXISTS idx_doi           ON files(doi);
CREATE INDEX IF NOT EXISTS idx_file_type     ON files(file_type);
CREATE INDEX IF NOT EXISTS idx_source        ON files(source_name);
CREATE INDEX IF NOT EXISTS idx_matched_query ON files(matched_query);
```